

Markovian and multi-curve friendly parametrisation of a HJM model used in valuation adjustment of interest rate derivatives.

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Motivation

XVA NEEDS A JOINT, SIMULABLE CURVE MODEL

- Valuation adjustments (CVA, DVA, FVA and related XVA terms) on interest-rate derivative portfolios require simulating the entire curve system, discounting and projection curves jointly, over long horizons and across collateralised and uncollateralised netting sets.
- The Heath–Jarrow–Morton framework prices consistently by construction, but a generic HJM volatility makes the forward-curve state non-Markovian, which is computationally prohibitive in an XVA engine.
- The paper constructs a parametrisation that is simultaneously Markovian and consistent with the post-crisis multi-curve setting.

The HJM framework and the Cheyette subclass

MODEL CLASS

Under the risk-neutral measure the instantaneous forward curve follows

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T) dW_t, \quad \alpha(t, T) = \sigma(t, T) \int_t^T \sigma(t, s) ds,$$

with the drift pinned down by no-arbitrage.

- The paper demonstrates the importance of the Cheyette subclass, separable volatility structures that collapse the curve dynamics onto a finite-dimensional Markovian state, and derives the instantaneous forward-rate dynamics of that form generically.
- Markovianity is what makes the specification implementable inside an XVA engine, where the whole curve system is simulated over long horizons.

Multi-curve toolkit

POST-CRISIS MARKET STRUCTURE

- After the financial crisis, IBOR-based deposit rates and OIS rates diverged abruptly, from a plateau of a few basis points to more than 200 bp at the peak, and FRA par rates detached from the corresponding forwards: discounting and projection require separate curves.
- The paper develops the multi-curve construction and its XVA integration, with pricing formulae for the vanilla non-linear building blocks (swaptions, caplets and floorlets) in the multi-curve setting.
- The focus is a relatively less liquid market, the Polish PLN curve system, and the calibration problems that illiquidity creates.

Calibration results and implications

ONE-FACTOR CHEYETTE MODELS ON PLN DATA

- Data availability disciplines the model choice: with swaption volatilities only, and no active caps and floors market, a three-factor exponential specification cannot be calibrated on the PLN market.
- Calibrations of several one-factor Cheyette-class models show that a relatively simple specification, Hull–White with two summands, achieves satisfactory calibration quality and computation time.
- Practical guidance for the XVA engine: the calibration step can be scheduled ahead of scenario generation, and the XVA integral should be discretised no finer than the simulation's horizon steps; overall performance depends on many degrees of freedom beyond the model choice (time-scale granularity, interpolation, minimisation method).

Thank you.

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